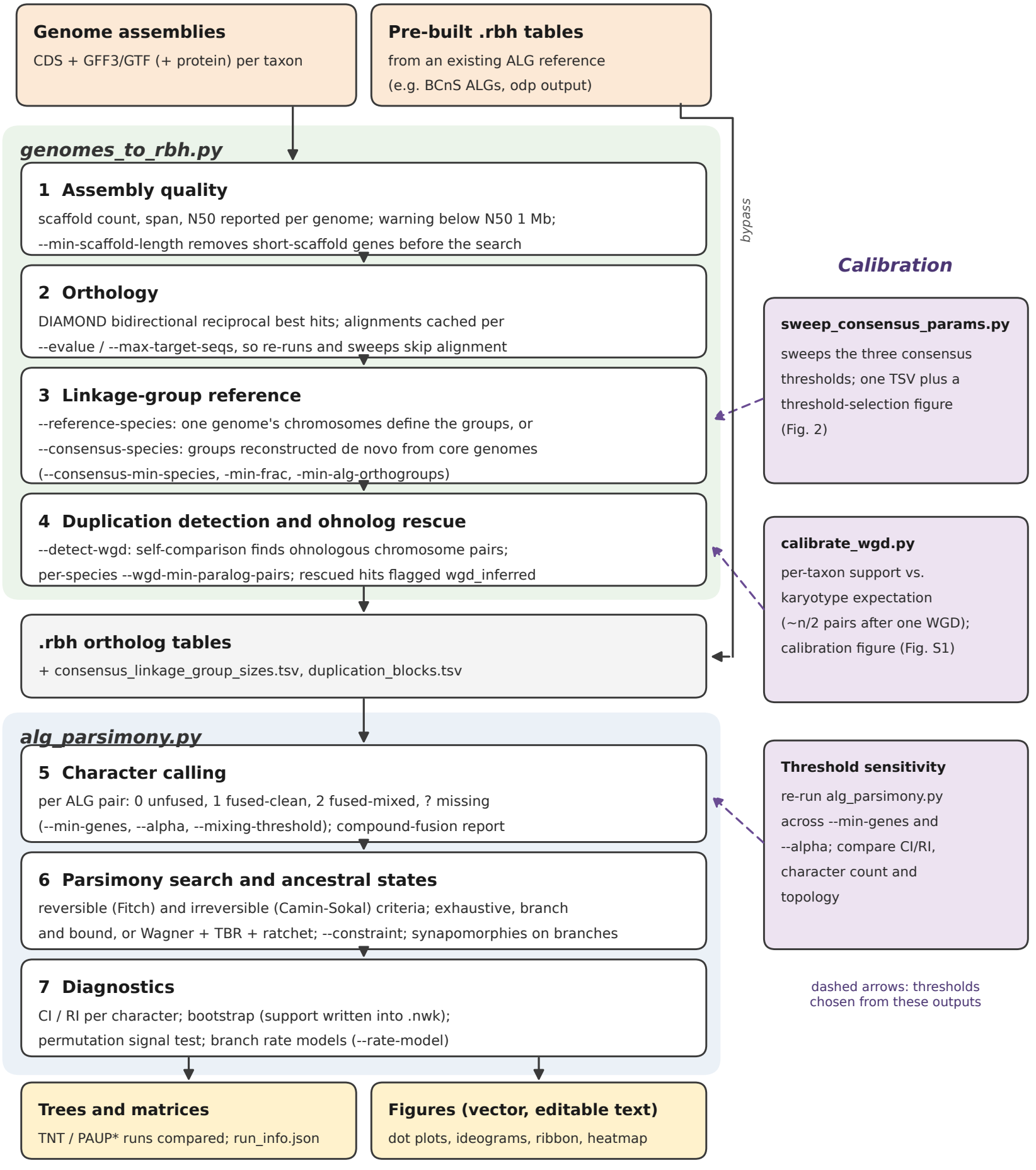
Genome assemblies
CDS + GFF3/GTF (+ protein) per taxon
Pre-built .rbh tables
from an existing ALG reference
(e.g. BCnS ALGs, odp output)
genomes_to_rbh.py
1 Assembly quality
scaffold count, span, N50 reported per genome; warning below N50 1 Mb;
--min-scaffold-length removes short-scaffold genes before the search
2 Orthology
DIAMOND bidirectional reciprocal best hits; alignments cached per
--evalue / --max-target-seqs, so re-runs and sweeps skip alignment
3 Linkage-group reference
--reference-species: one genome's chromosomes define the groups, or
--consensus-species: groups reconstructed de novo from core genomes
(--consensus-min-species, -min-frac, -min-alg-orthogroups)
4 Duplication detection and ohnolog rescue
--detect-wgd: self-comparison finds ohnologous chromosome pairs;
per-species --wgd-min-paralog-pairs; rescued hits flagged wgd_inferred
bypass
.rbh ortholog tables
+ consensus_linkage_group_sizes.tsv, duplication_blocks.tsv
Calibration
sweep_consensus_params.py
sweeps the three consensus
thresholds; one TSV plus a
threshold-selection figure
(Fig. 2)
calibrate_wgd.py
per-taxon support vs.
karyotype expectation
(~n/2 pairs after one WGD);
calibration figure (Fig. S1)
alg_parsimony.py
5 Character calling
per ALG pair: 0 unfused, 1 fused-clean, 2 fused-mixed, ? missing
(--min-genes, --alpha, --mixing-threshold); compound-fusion report
6 Parsimony search and ancestral states
reversible (Fitch) and irreversible (Camin-Sokal) criteria; exhaustive, branch
and bound, or Wagner + TBR + ratchet; --constraint; synapomorphies on branches
7 Diagnostics
CI / RI per character; bootstrap (support written into .nwk);
permutation signal test; branch rate models (--rate-model)
Threshold sensitivity
re-run alg_parsimony.py
across --min-genes and
--alpha; compare CI/RI,
character count and
topology
dashed arrows: thresholds
chosen from these outputs
Trees and matrices
TNT / PAUP* runs compared; run_info.json
Figures (vector, editable text)
dot plots, ideograms, ribbon, heatmap